

EE 3201.302

Lab 6 DC Electric Motors

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I. Abstract

This experiment investigates the operational characteristics of a DC electric motor, with a specific focus on its torque-speed relationship and efficiency. Using a P711A hobby motor, the no-load rotational speed and torque at various input voltages were measured to construct a torque vs. speed curve. Additionally, the experiment aimed to determine the motor's maximum mechanical power output and efficiency at two different input voltages. The results offer valuable insights into the motor's behavior under different load conditions, highlighting its capacity to convert electrical power into mechanical motion and its overall performance in real-world applications. These findings contribute to a deeper understanding of the motor's efficiency and operational limits.

II. Introduction

DC electric motors are essential devices that convert electrical power into mechanical energy, playing a crucial role in a wide range of applications, from small household appliances to large industrial machinery. These motors are valued for their simplicity, efficiency, and versatility in various systems. In this experiment, the primary focus is on understanding the operational characteristics of a DC motor, specifically by analyzing its torque-speed relationship. By investigating how different input voltages impact the motor's performance, the experiment aims to assess key parameters, including its no-load speed, stall torque, mechanical power output, and efficiency under varying conditions. This analysis will contribute to a better understanding of the motor's behavior, enabling improvements in motor design and optimization for various applications.

III. Procedure

1. DC Motor Setup

- a. Obtain a P711A hobby motor and a plastic wheel.
 - i. Make sure that all but one of the spokes of the plastic wheel are non-reflecting.
 - ii. Securely connect the wheel to the shaft and make sure that it is not tilted.
- b. Examine the DC motor.
 - i. Identify its positive and negative terminals.
 - ii. Re-read the motor's data sheet carefully and pay special attention to its no-load rotational speed at the rated voltage.
- c. Mount the motor onto the stand securely, but without squeezing the motor too tightly. Squeezing the motor too tightly will add friction on the rotor and can cause your measurements to be incorrect.
- d. Turn on the DC power supply and make sure that the +25 V output is set to source 0 V and is not yet enabled, voltage to the motor should not surpass 9V.
 - i. Connect the DC power supply, DC voltmeter and DC ammeter to the terminals of the motor.

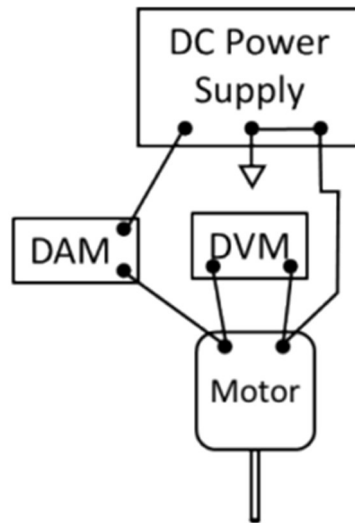


Figure 3.1: Electrical Connections for the first tests. The Digital Ammeter (DAM – Keysight 34401) is connected in series with the motor and the Digital Voltmeter (DVM – Keysight 34401) is connected in parallel. The 25 V output of the DC power supply is connected in parallel too.

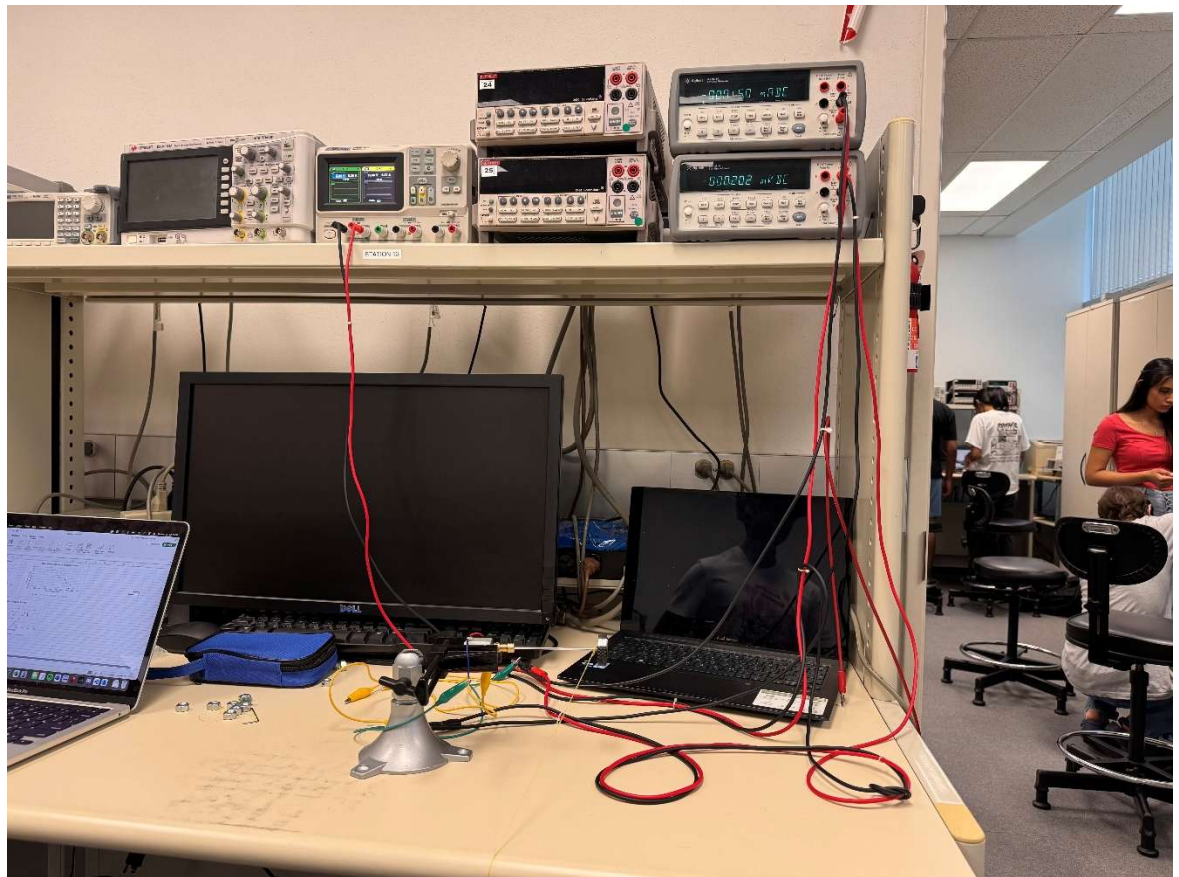


Figure 3.2: DC Motor setup corresponding to diagram from Figure 3.1.

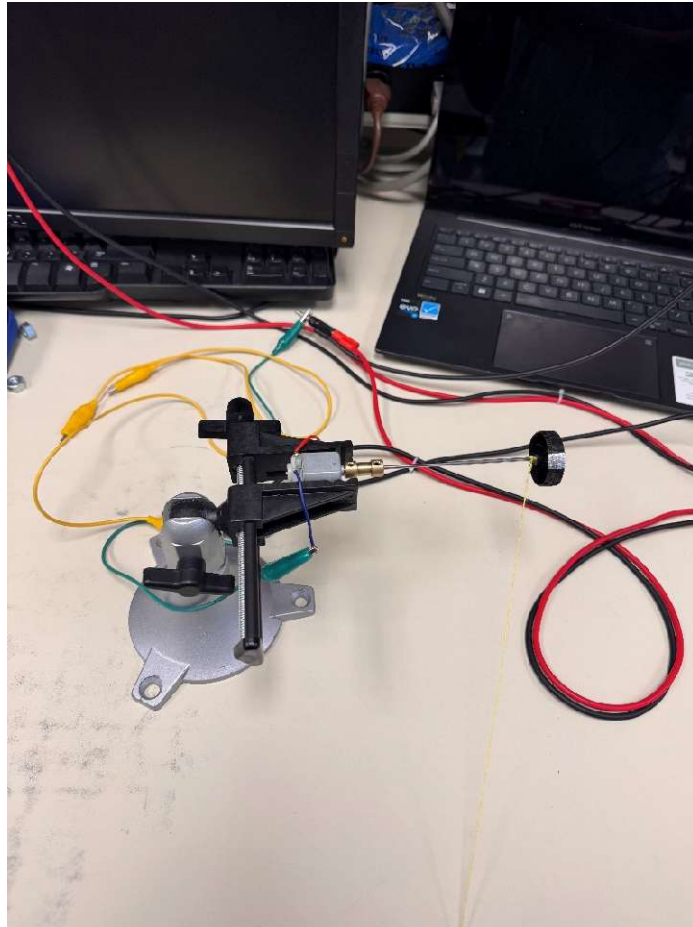


Figure 3.3: DC Motor mounted on the stand and connected to DAM, DVM, and power supply.

2. Measuring the “No-Load Speed” for the motor.
 - a. Set the power supply to 2 V and use the tachometer to measure the approximate no load speed of the motor.
 - b. Repeat for at least 5 different input voltages between 2 and 9 Volts (2, 4, 5, 7, and 9).
 - c. When finished, you should have a table of data with four columns: Voltage (V), Current (A) as well as Rotational Speed (RPM). Add three more columns to the table: Electrical Power (Watts) which is simply $\text{Voltage} * \text{Current}$, and ii. Rotational Speed in Rotations / second (RPS) which is simply $\text{RPM}/60$.
 - d. Make a quick plot of your data. Plot Max-RPM versus Voltage and do a fit to the data in MS Excel.

	A	B	C	D	E	F
1		2 Volts	4 Volts	5 Volts	7 volts	9 volts
2	Voltage	1.79	3.56	4.9	6.85	8.92
3	Current	36.9	78	59.6	88.8	0.14
4	No load speed	2378 RPM	3489 RPM	3679 RPM	5286 RPM	7206 RPM
5						
6	Electrical Power(W)	66.051	277.68	292.04	608.28	1.2488
7	Rotational Speed (RPS)	39.63	58.15	61.32	88.1	120.1
8	Radians per second	248.88	365.182	385.09	553.27	754.23

Figure 3.4: Data of measured voltage, current, and RPM and different voltage levels. Calculations of Electrical Power, RPS, and Rad/s.

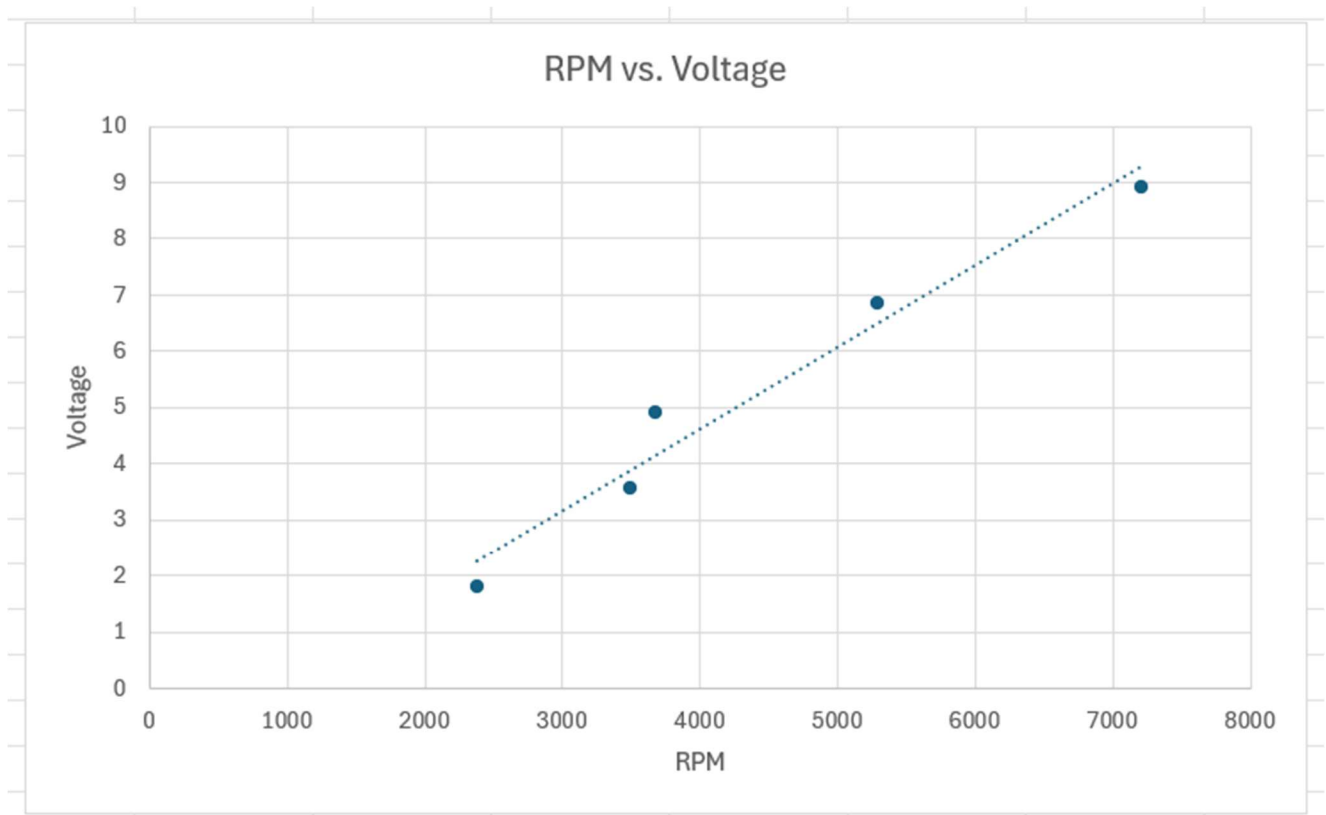


Figure 3.5: Graph of RPM vs. Voltage from values gathered in the experiment.

3. Measuring the torque vs rotational speed curves for the motor.
 - a. Attach a string on the shaft as demonstrated. Attach a wheel at the end of the spindle.
 - b. Perform Experiment and Record Data
 - i. We are going to attach weights (washers) at the end of the string and measure the rise time of the weights from the ground to the level of your lab table. Each washer is approximately 5.2 grams.
 - ii. With your cell phone stopwatch, measure the rise time of the weights from the ground level to your table level. Record the Voltage, the rise time and the current readings.
 - iii. Add or remove weights from the end of the string and repeat the experiment, recording the voltage, raise time and the ammeter readings each time. Be sure to put the weights on in a random order! Do not simply add weight each time. Instead, choose the weight to use in a random order. This helps to prevent systematic errors in the experiment.
 - iv. To increase the accuracy of the experiment, repeat the experiment with each weight at least 2 times and preferably 3 times. Obtain the average values for both the rise time and current reading from the ammeter.
 - v. Enter your data into a table repeat your measurements for both of your chosen voltage settings.

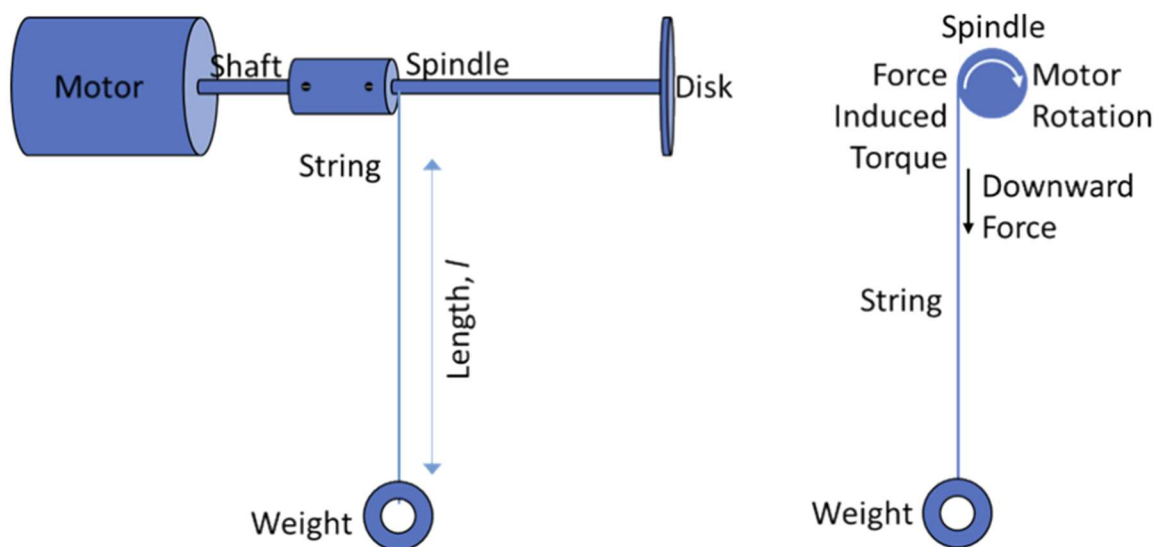


Figure 3.6: The motor with an attached spindle and string to wind up a weight. (Left) Schematic diagram of the setup showing the spindle attached to the motor shaft, the string, and the weight. (Right) Diagram of the force induced on the motor spindle by the weight.

4. Data Analysis

- a. The motor torque versus rotational speed curve at each voltage setting.

If torque is high, then rotational speed will be less. If rotational speed is high, then torque will be lower. Our data points are not in an exact linear line, but it is expected because real life experiments have errors and calibration differences. The best fit line matches both voltage levels, making the graph correct.

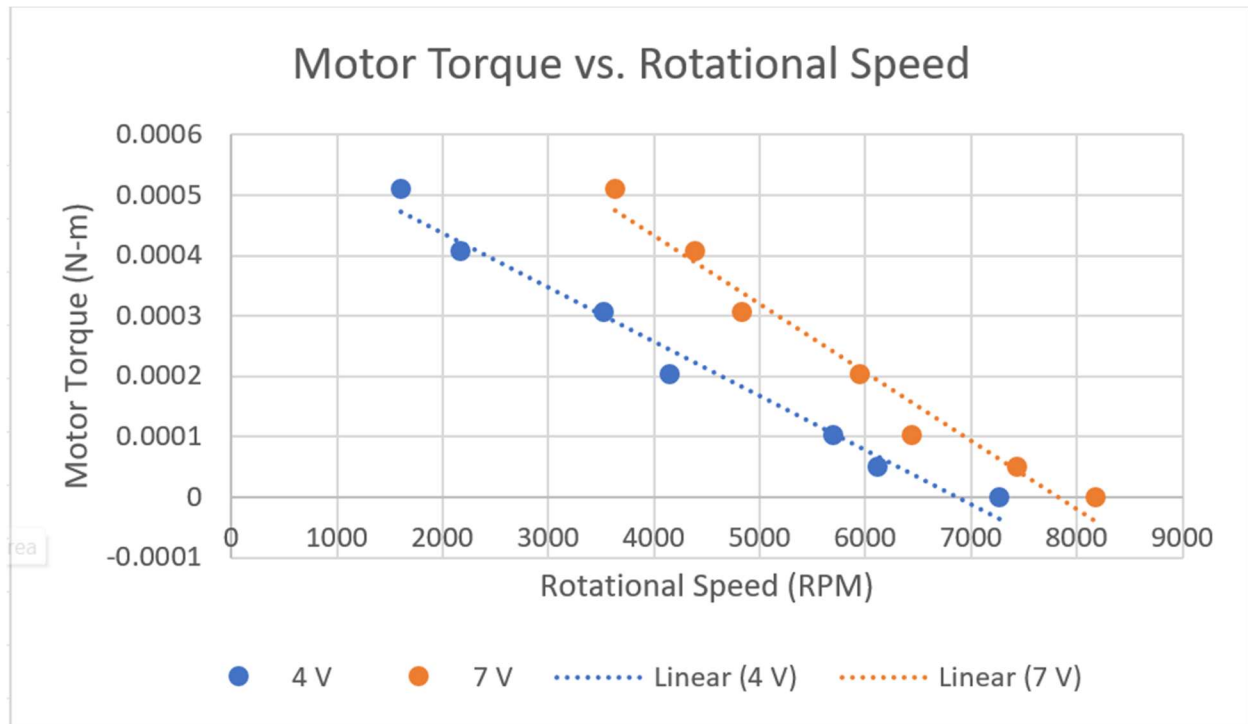


Figure 3.9: Motor Torque vs. Rotational Speed Graph

- b. The electrical power entering the motor as a function of the rotational speed at each voltage.

The power entering the motor is proportional to the power leaving the motor since we are converting electrical power into mechanical power. The speed and torque of the motor is dependent on the input of electrical power. Our graph shows a slight decrease in electrical power as the rotational speed increases. Electrical power is somewhat constant for each voltage level but tends to be higher when the motor rotates at lower rotational speeds. The graph is consistent with our knowledge and the reason for the slight decrease may be due to resistive loss.

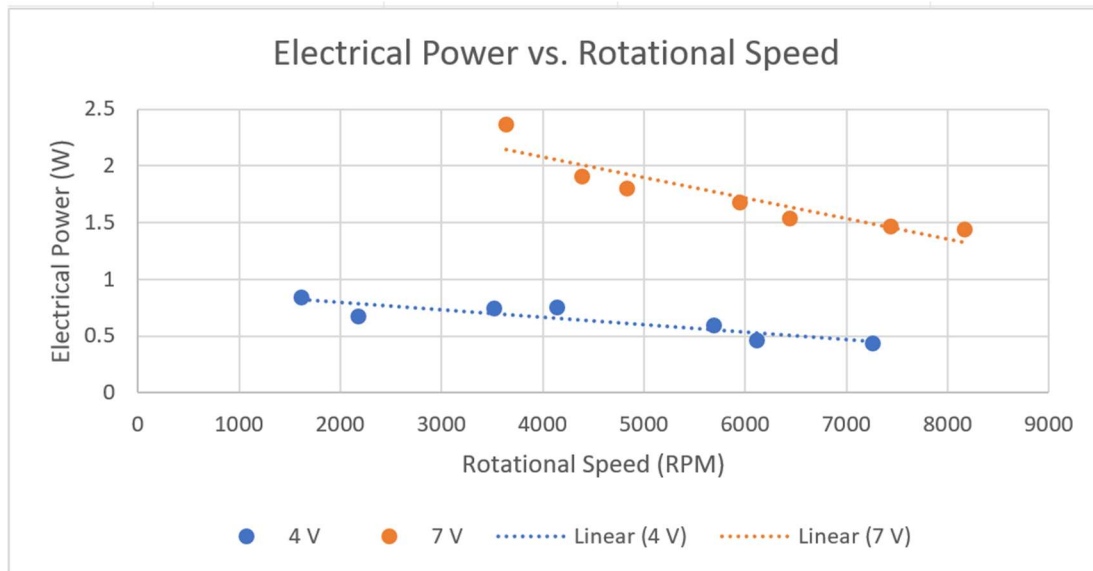


Figure 3.10: Electrical Power vs. Rotational Speed Graph

- c. The mechanical power exerted by the motor as a function of the rotational speed at each voltage.

The mechanical power of the motor depends on rotational speed and torque, such as $P = \omega * t$. Therefore, mechanical power should increase with voltage and speed. However, in our graph, power decreases with speed, likely due to factors like increasing back EMF, frictional losses, or motor saturation. This behavior is common in DC motors, as shown by the initial increase in power followed by a decrease. The 4-volt experiment appears more accurate than the 7-volt experiment, as it operates at lower speeds, providing greater accuracy in the results.

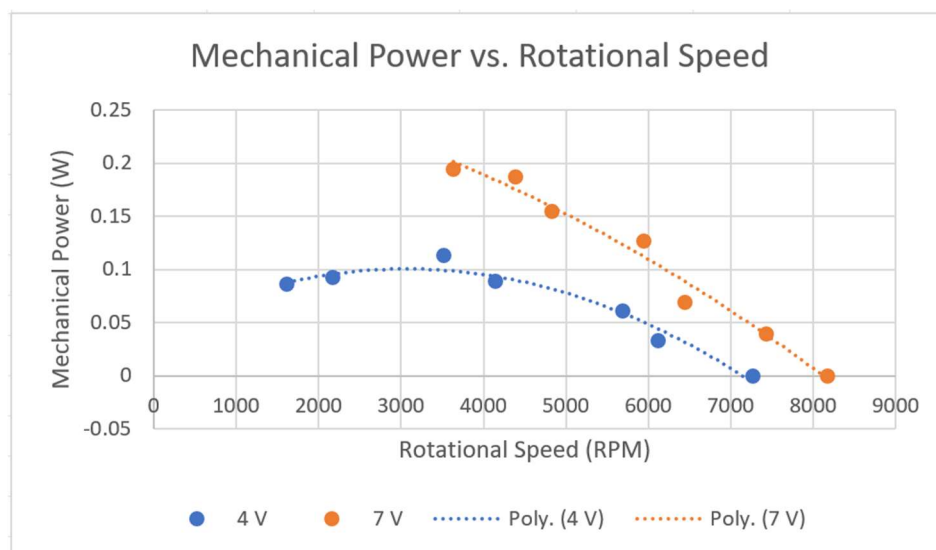


Figure 3.11: Mechanical Power vs. Rotational Speed Graph

- d. Motor power efficiency is a function of the rotational speed at each voltage.

Motor power efficiency is dependent on the input of electrical power and the output mechanical power ratio. The peak efficiency should be when torque and rotational speed are at half the max value. In our graph below, we see the 4-volt line is exactly what we expected because the efficiency peaks at about half the max rotational speed. The 7-volt line has a very small peak at about 4300 RPM. The data for the 7-volt part is not incorrect, but the reason why the graph is not as expected is because of other factors with the motor like saturation, EMF loss, and resistance. We saw this issue in the other graph as well and it makes sense. This graph shows the results better at a lower voltage.

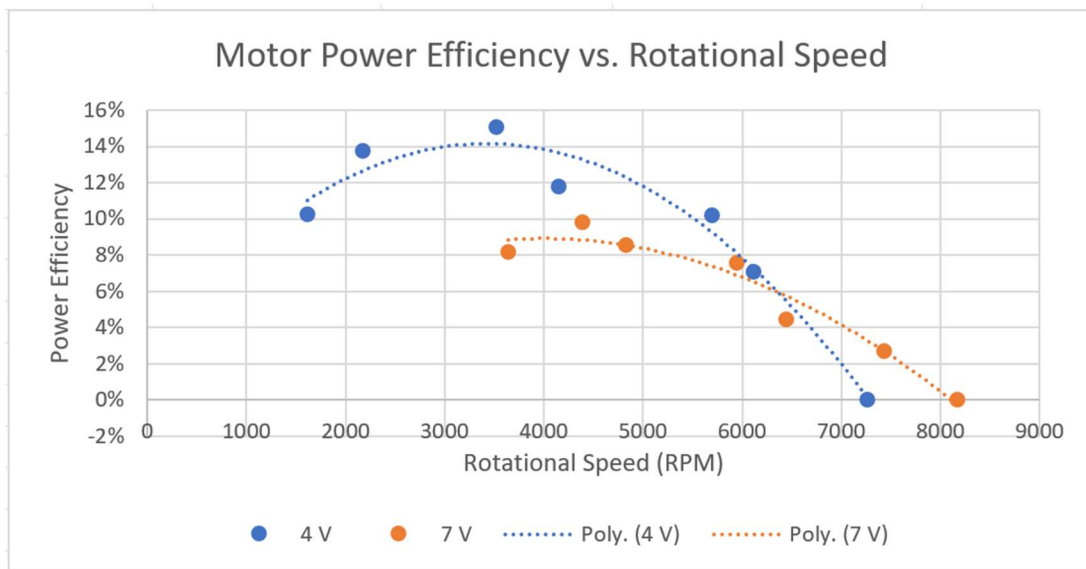


Figure 3.12: Motor Power Efficiency vs. Rotational Speed Graph

- e. The torque versus motor current curve at each voltage.

The torque and current relationship should be positive linear with a constant. Since $t * \omega = (v * I) (e)$. Let (e) be an efficiency constant. If we rearrange the equation, we can let ω and v be a constant too then we get: $t = K * I$. Therefore, we will have a linear graph. This is consistent with our findings and as the voltage increases so will the current and then the torque and speed will increase as well. The graph below shows a linear line for the 4 volt which is as expected. However, for the 7-volt line it is linear then starts to flatten out at the end this is again due to the motors conditions we discussed in the previous two graphs. Therefore the 4 volt is a better representation and operating motors at higher voltages may cause differences in data.

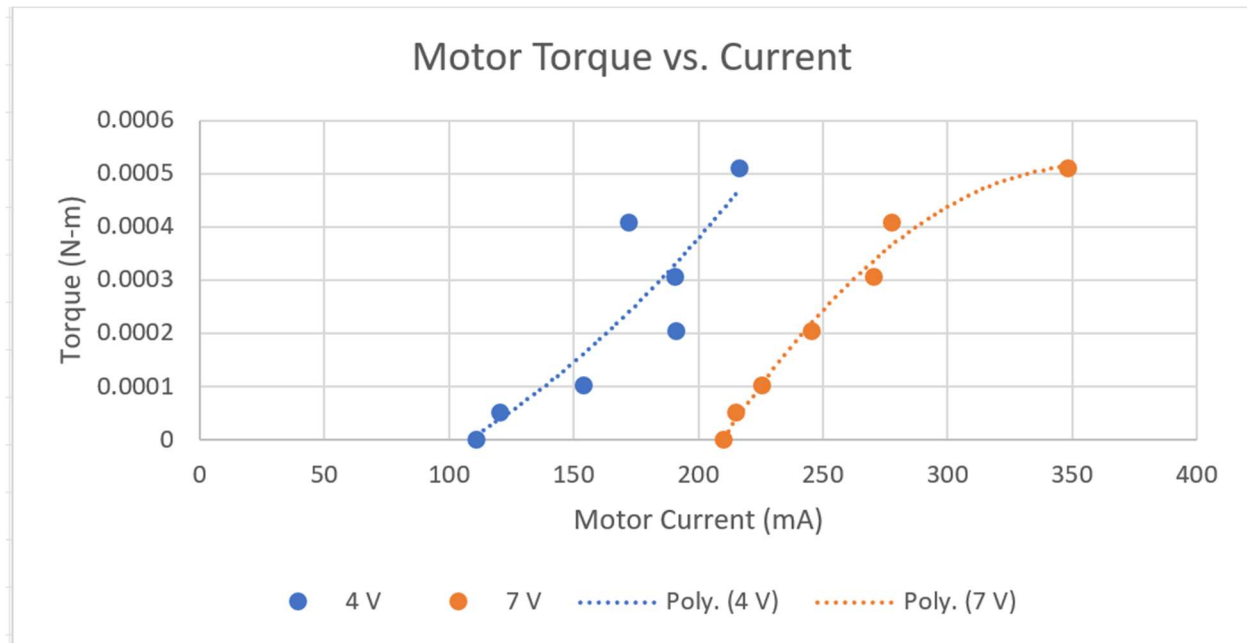


Figure 3.13: Motor Torque vs. Current Graph

IV. Experimental Results

1. The no load speed of the motor in RPM as we increased the input voltage.

As we increase the input voltage from 2V to 9V, the no-load rotational speed (RPM) of the motor also increases, which is consistent with the motor's typical behavior since increasing input power will increase output power. The motor's speed is directly proportional to the input voltage, demonstrating a linear relationship. At higher voltages, the motor overcomes internal resistance more efficiently, resulting in faster rotation. The data points plotted are not 100% accurate because there are calibration tolerances within the DC motor, power supply, and other resistances. The tachometer could be another factor for our data points not being exact. As we were measuring the RPM the value we got fluctuated and we had to make our best judgement. Overall, the data was in a reasonable range of what was expected, and we created a linear best fit line which matches our theoretical understanding of DC motors if we increase voltage.

2. Measuring Torque vs. Rotational speed.

The torque vs. rotational speed curve showed that as torque increases, the rotational speed decreases. This is in accordance with the expected behavior for a DC motor, where a higher load (torque) results in a slower speed. Otherwise, no load (torque) results in higher rotation speeds. The data points for torque and speed were not perfectly linear due to experimental errors and calibration differences, but the general trend was consistent. As we were doing the experiment we had to ensure that we went in a random order to avoid systematic

errors and that gave us accurate data. The experiment made sense because as we added more load on to the motor it took more time to reach the top. After the experiment for each voltage, we were able to analyze the data. The graphs gave us clarity on different measurements of our motor, like power, rotation speed, and efficiency. The data was mostly consistent with what we expected and gave us a better understanding of what happened in the experiment.

V. Discussion and Conclusions

The results of this experiment provide valuable insights into the operational characteristics of a DC motor, focusing on its torque-speed relationship, efficiency, and power output at varying voltages. As expected, the no-load speed of the motor increased with higher input voltage. This behavior is consistent with the fundamental principles of DC motors, where an increase in voltage leads to more current flow through the motor, resulting in a higher rotational speed. This relationship was confirmed by the data, which showed a clear, linear increase in speed as the voltage was raised from 2V to 9V.

The torque-speed curves obtained during the experiment revealed an inverse relationship between torque and rotational speed. This behavior is typical for DC motors: as the motor produces more torque to lift heavier loads, the speed decreases. The torque-speed data for both 4V and 7V settings exhibited this trend, with higher voltages leading to higher torque but lower speeds. While the data was not perfectly linear due to experimental errors and calibration differences, the general trend was consistent with theoretical expectations.

Regarding mechanical power and efficiency, the results showed that mechanical power initially increased with rotational speed but started to decrease at higher speeds. This decrease in mechanical power at high speeds can be attributed to factors such as increasing back EMF, frictional losses, and motor saturation. The 4V setting produced more accurate results due to lower speeds, which minimized the impact of these losses. The motor's efficiency also followed a predictable pattern, peaking at mid-range speeds. The 4V setting showed the highest efficiency, while the 7V setting demonstrated reduced efficiency at higher speeds due to additional losses in the motor.

In conclusion, this experiment highlighted the importance of input voltage in determining motor performance. Operating the motor at lower voltages provided more consistent and reliable data, while higher voltages led to additional losses that affected the accuracy of the results. These findings contribute to a deeper understanding of DC motor behavior, which can be used to optimize motor design and efficiency for various real-world applications.